

Virtual Formula 2021

Rules & Regulations

Contents

1. Competition Overview	2
1.1 Description.....	2
1.1 Objective.....	2
1.2 Classes	2
1.3 Events	2
2. Vehicle Model Optimization Rules.....	2
2.1. Legal Parameters	2
2.2. Illegal Parameters	4
2.3 Simulink	4
3. Events	4
3.1. Skidpad	4
3.2. Acceleration.....	5
3.3. Autocross	5
3.4. Endurance.....	6
3.5. Efficiency CV	6
3.6. Efficiency EV.....	7
3.7. Penalties	7
4. Submissions	7
4.1. Requirements	7
4.2. Late Submissions	8

1. Competition Overview

The following section describes the competition in brief.

1.1 Description

The competition is influenced by official formula student competitions. The virtual vehicle model and events are based on their real-world counterparts. Aspects of the competition have been adjusted to better fit the virtual world.

1.1 Objective

To challenge teams to optimize a rule compliant virtual vehicle model to achieve the maximum score across the scored events.

1.2 Classes

The competition is separated into two classes (Internal Combustion Engine (CV) and Electric Vehicle (EV)).

1.3 Events

The following table describes the objective of the competition events.

Event	Objective
Skidpad	Minimize time to complete the course.
Acceleration	Minimize time to complete the course.
Autocross	Minimize time to complete the course.
Endurance	Minimize time to complete the course.
Efficiency	Maximize energy efficiency.

Table 1: Event Objectives

The following table describes the maximum possible score for each of the competition events.

Event	Maximum Score
Skidpad	75
Acceleration	75
Autocross	100
Endurance	325
Efficiency	100

Table 2: Event Maximum Scores

2. Vehicle Model Optimization Rules

The following section describes how competitors may optimize the vehicle model.

2.1. Legal Parameters

Competitors may optimize the vehicle model parameters described in the following tables, where:

- Table 3 is applicable to both CV and EV classes.
- Table 4 is applicable for EV class only.
- Table 5 is applicable for CV class only.

Subsystem	Subtab	Tunable parameters	Min	Max	Notes
Body	Sprung Mass	Wheelbase	1525	2500	
		cg longitudinal front wheel	0	Wheelbase	
	Skidplate FrontLeft	Loc Y	N/A	N/A	(front suspension track width)/2
		Loc Z	N/A	N/A	Front Left Ride Height
	Skidplate FrontRight	Loc Y	N/A	N/A	-(front suspension track width)/2
		Loc Z	N/A	N/A	Front Right Ride Height
	Skidplate RearLeft	Loc X	N/A	N/A	-wheelbase
		Loc Y	N/A	N/A	(front suspension track width)/2
		Loc Z	N/A	N/A	Rear Ride Height
	Skidplate RearRight	Loc X	N/A	N/A	-wheelbase
		Loc Y	N/A	N/A	-(front suspension track width)/2
		Loc Z	N/A	N/A	Rear Ride Height
	Aerodynamic Forces	Property File	N/A	N/A	Choose from provided aero file options (files must not be edited).
		Front Downforce Location	N/A	N/A	Z = Front Ride Height
		Rear Downforce Location	N/A	N/A	X = -wheelbase Z = Rear Ride Height
		Drag Force Location	N/A	N/A	X = -(cg longitudinal front wheel distance)
		Front Sensor Location	N/A	N/A	Z = Front Ride Height
		Rear Sensor Location	N/A	N/A	X = -wheelbase Z = Rear Ride Height
		Front Downforce Scale	0	2	
		Rear Downforce Scale	N/A	N/A	2-(Front Downforce Scale)
Brakes	Parameters	Bias Front	0	1	
		Master cylinder pressure gain	0	1	
Front Suspension	Wheel Location	Track width	1000	2000	>=75% rear track width
	Suspension Setup Data	Ride Height Adjustment Active	N/A	N/A	Must be activated
		Left Ride Height Sensor Ride Height	0	inf	
		Left Ride Height Sensor Locations	N/A	N/A	X = 0 Y = (front suspension track width)/2 Z = Front Left Ride Height
		Right Ride Height Sensor Ride Height	N/A	N/A	= Left Ride Height Sensor Ride Height
		Right Ride Height Sensor Locations	N/A	N/A	X = 0 Y = -(front suspension track width)/2 Z = Front Right Ride Height
		Wheel Angles Toe and Camber	-inf	inf	
	All others	All	N/A	N/A	Complete freedom
Rear Suspension	Wheel Location	Track width	1000	2000	>=75% front track width
	Suspension Setup Data	Ride Height Adjustment Active	N/A	N/A	Must be activated
		Center Ride Height Sensor Ride Height	0	inf	
		Center Ride Height Sensor Locations	N/A	N/A	X = -wheelbase Y = 0 Z = Rear Ride Height
		Wheel Angles Toe and Camber	-inf	inf	
	All others	All	N/A	N/A	Total freedom
Steering	All	All	N/A	N/A	Total freedom

Front Wheels	Wheels	Tire Property File	N/A	N/A	Choose from provided tire file options (files must not be edited).
Rear Wheels	Wheels	Tire Property File	N/A	N/A	Choose from provided tire file options (files must not be edited).

Table 3: General Tunable Parameters

Subsystem	Subtab	Tunable parameters	Min	Max	Notes
Powertrain	Motors	Torque Scaling	0	1	
		Transmission Ratio	0	inf	

Table 4: EV Specific Tunable Parameters

Subsystem	Subtab	Tunable parameters	Min	Max	Notes
Powertrain	Engine	Transmission Ratio	0	inf	
	Transmission	Gear Ratios	N/A	N/A	Total Freedom
		Gear Shift Method	N/A	N/A	Total Freedom
		Upshift RPM	0	inf	
		Downshift RPM	0	inf	
	Differentials	Inertia	N/A	N/A	Must not be tuned
		Reaction Torques	N/A	N/A	Must not be tuned
		All others	N/A	N/A	Total Freedom

Table 5: CV Specific Tunable Parameters

2.2. Illegal Parameters

Any parameter not described in the previous tables must not be adjusted. Adjusting an illegal parameter will disqualify the competitor from all scored events.

2.3 Simulink

Competitors must only edit the provided Simulink models within the green subsystem blocks.

3. Events

The following section describes the scored events. The same vehicle model and setup must be used across all events.

3.1. Skidpad

The skidpad course consists of two pairs of concentric circles in a figure of eight pattern. The circles have a radius of 9.125m measured from the centreline of the road. The road has a constant width of 3m. At the outside edges of the road, there is a kerb of 100mm height that is intended to penalize vehicles exploiting track limits.

The following event settings must be set:

- Path File: VF2020_Skidpad.drd
- Road Data File: VF2020_Skidpad.rdf
- Road Graphic File: VF2020_Skidpad.obj
- Flying lap: off
- User Defined Location: on (0,0, Automatic Z Location on,0,0,0)
- Laps: on, min = 1, max = 1

The following equation describes how the skidpad score is determined.

$$SkidpadScore = 75 \times \left(\frac{\left(\frac{T_{max}}{T_{team}} \right)^2 - 1}{0.5625} \right)$$

Where:

- T_{team} is the time to complete the course
- T_{max} is the best time of all the teams in the class multiplied by 1.25
- Minimum score is 0

3.2. Acceleration

The acceleration course is a straight line with length of 75m.

The following event settings must be set:

- VI Driver Input File: VF2020_Acceleration.vdf
- Road Data File: VF2020_Acceleration.rdf
- Road Graphic File: VF2020_Acceleration.obj
- User Defined Location: on (0,0,Automatic Z Location on,0,0,0)

The following equation describes how the acceleration score is determined.

$$AccelerationScore = 75 \times \left(\frac{\frac{T_{max}}{T_{team}} - 1}{0.5} \right)$$

Where:

- T_{team} is the time to complete the course
- T_{max} is the best time of all the teams in the class multiplied by 1.5
- Minimum score is 0

3.3. Autocross

The autocross course is a closed circuit. The road has a constant width of 3m. At the outside edges of the road, there is a kerb of 100mm height that is intended to penalize vehicles exploiting track limits.

The following event settings must be set:

- Path File: VF2020_RaceTrack.drd or a .drd file created using VI-Road's corner cutting tool.
- Road Data File: VF2020_RaceTrack.rdf
- Road Graphic File: VF2020_RaceTrack.obj
- Flying lap: off
- Initial Velocity: 1000
- User Defined Location: on (0,0,Automatic Z Location on,0,0,0)
- Laps: on, min = 1, max = 1

The following equation describes how the autocross score is determined.

$$AutocrossScore = 100 \times \left(\frac{\frac{T_{max}}{T_{team}} - 1}{0.25} \right)$$

Where:

- T_{team} is the time to complete the course
- T_{max} is the best time of all the teams in the class multiplied by 1.25
- Minimum score is 0

3.4. Endurance

The endurance course is a closed circuit. The road has a constant width of 3m. At the outside edges of the road, there is a kerb of 100mm height that is intended to penalize vehicles exploiting track limits.

The following event settings must be set:

- Path File: VF2020_RaceTrack.drd or a .drd file created using VI-Road's corner cutting tool.
- Road Data File: VF2020_RaceTrack.rdf
- Road Graphic File: VF2020_RaceTrack.obj
- Flying lap: on
- User Defined Location: on (0,0, Automatic Z Location on, 0, 0, 0)
- Laps: on, min = 1, max = 1

The following equation describes how the endurance score is determined.

$$EnduranceScore = 325 \times \left(\frac{\frac{T_{max}}{T_{team}} - 1}{0.333} \right)$$

Where:

- T_{team} is the time to complete the course
- T_{max} is the best time of all the teams in the class multiplied by 1.333
- Minimum score is 0

3.5. Efficiency CV

Energy efficiency is measured during the endurance event. Teams who score zero for endurance will also score 0 for efficiency.

The following equation describes how the efficiency score is determined.

$$EfficiencyScore = 100 \times \left(\frac{\frac{E_{min}}{E_{team}} - 1}{\frac{E_{min}}{E_{max}} - 1} \right)$$

Where:

- E_{team} is the team's efficiency factor
- E_{max} is the highest efficiency factor of all teams
- E_{min} is the lowest efficiency factor of all teams

The following equation describes how the efficiency factor is determined.

$$EfficiencyFactor = \left(\frac{T_{min} \times V_{min}}{T_{team} \times V_{team}} \right)$$

Where:

- T_{team} is the team's endurance time
- T_{min} is the fastest endurance time of all teams
- V_{team} is the team's total fuel consumption from the endurance event
- V_{min} is the lowest total fuel consumption from the endurance event of all teams

3.6. Efficiency EV

Energy efficiency is measured during the endurance event. Teams who score zero for endurance will also score 0 for efficiency.

The following equation describes how the efficiency score is determined.

$$EfficiencyScore = 100 \times \left(\frac{\frac{0.1}{E_{team}} - 1}{\frac{0.1}{E_{max}} - 1} \right)$$

Where:

- E_{team} is the team's efficiency factor
- E_{max} is the highest efficiency factor of all teams

The following equation describes how the efficiency factor is determined.

$$EfficiencyFactor = \left(\frac{T_{min} \times EN_{min}^2}{T_{team} \times EN_{team}^2} \right)$$

Where:

- T_{team} is the team's endurance time
- T_{min} is the fastest endurance time of all teams
- EN_{team} is the team's total energy used from the endurance event
- EN_{min} is the lowest total energy used from the endurance event of all teams

3.7. Penalties

- Off-course (all 4 wheels outside the track) will result in disqualification from the event

4. Submissions

The following section describes requirements for submissions and penalties or late submissions.

4.1. Requirements

The following is required for submission:

- Vehicle model database
- Fingerprint file for the events as they were run
- Simulink models and Matlab Scripts if used to run simulations
- Completed submission form

4.2. Late Submissions

Submissions later than the deadline will be penalised by ten points for every 24hours they are overdue. These points will be deducted from the team's overall score.